

Ivan Ricardo

PhD Candidate in Econometrics

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Citizenship: USA

EDUCATION

2023 – present	PhD in Econometrics Maastricht University · Expected completion: 2027 Thesis: <i>Tensor Methods in Econometrics</i> Advisors: Alain Hecq and Ines Wilms	Maastricht, NL
2021 – 2023	MSc in Economic and Financial Research Maastricht University · Specialization in Econometrics	Maastricht, NL
2017 – 2020	BA in Economics University of California, Santa Barbara	Santa Barbara, CA

RESEARCH INTERESTS

Time series econometrics, matrix- and tensor-valued time series, dimension reduction

JOB MARKET PAPER

Impulse Response Inference for Matrix Time Series

with A. Hecq and I. Wilms

Abstract: Matrix autoregressive (MAR) models offer a parsimonious framework for matrix-valued time series, yet tools for quantifying uncertainty around their impulse responses are lacking. We develop asymptotic and bootstrap-based inference for impulse responses of stable $MAR(p)$ models. We derive the joint asymptotic distribution of the coefficient and covariance estimators, which yields closed-form delta-method standard errors. To address finite-sample bias, we propose ProBAB-MAR, a bias-corrected bootstrap that projects the corrected coefficients back onto the Kronecker parameter space, and prove that it attains asymptotically correct coverage. Monte Carlo simulations show that delta-method intervals undercover in small samples, while ProBAB-MAR achieves near-nominal coverage with intervals considerably narrower than those from an unrestricted VAR. An application to Euro-area inflation illustrates how shocks transmit across countries and inflation categories.

PUBLICATIONS

Forthcoming	Decomposing Co-movements in Matrix-Valued Time Series: A Pseudo-Structural Reduced Rank Approach A. Hecq, I. Ricardo, and I. Wilms <i>Econometrics and Statistics</i>	JOURNAL
Forthcoming	Performance of Radio-Based Detection for Operational Monitoring of M5+ Class Solar Flares J. van Ravenswaaij, I. Wilms, I. Ricardo, and M. Brchnelova <i>Journal of Space Weather and Space Climate</i>	ARXIV
2025	Detecting Cointegrating Relations in Non-Stationary Matrix-Valued Time Series A. Hecq, I. Ricardo, and I. Wilms <i>Economics Letters</i> , 248, 112205	JOURNAL

WORKING PAPERS

2024	Reduced Rank Matrix Autoregressive Models: A Medium N Approach A. Hecq, I. Ricardo, and I. Wilms <i>Working paper</i>	ARXIV
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PRESENTATIONS

2026	Netherlands Econometrics Study Group (poster), Midwest Econometrics Group (scheduled), CFE-CMStatistics* (scheduled)
2025	CFE-CMStatistics*
2024	CFE-CMStatistics*, Econometrics Seminar (Maastricht University), Netherlands Econometrics Study Group (poster)
	* Invited session

PROFESSIONAL ACTIVITIES

Referee	Journal of Econometrics
Workshops	Network Econometrics Summer School, Ca' Foscari University of Venice (2023) Dimensionality Reduction and Inference in High-Dimensional Time Series (2022)

TEACHING

Tutor	Maastricht University	Avg. student evaluation
	Econometric Methods I (BSc Econometrics & Operations Research)	8.60/10
	Macroeconomics (BSc Econometrics & Operations Research)	8.69/10
	Quantitative Methods III (BSc International Business)	8.35/10
	Statistics I (University College Maastricht)	4.7/5
Supervision	MSc thesis (Econometrics & Operations Research)	1 student
	BSc thesis (Econometrics & Operations Research)	4 students
	Topics include graphical models for matrix time series, clustering, and forecasting solar flares	

SOFTWARE

Julia	CommonFeatures.jl MatrixAutoRegressions.jl	GraphicalLasso.jl PseudoStructuralComovements.jl
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SKILLS

Programming	Julia, R, Python, MATLAB, Bash, Lua
Tools	Git, Docker, Linux, LaTeX
Languages	English and Spanish (native), Dutch (basic)

REFERENCES

Alain Hecq Professor Maastricht University a.hecq@maastrichtuniversity.nl	Ines Wilms Associate Professor Maastricht University i.wilms@maastrichtuniversity.nl	S. Yaser Samadi Associate Professor Southern Illinois University ysamadi@siu.edu
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